SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 20

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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives :

To perform database modification operations using JDBC.

Theory :

JDBC (Java Database Connectivity) enables Java applications to interact with relational databases & perform database operations using SQL statements. INSERT, UPDATE and DELETE are DML operations used to modify data stored. The INSERT statement is used to add new records, UPDATE statement modifies existing records, and DELETE statement removes records from a table. These can be performed using the **Statement or PreparedStatement** interface.

The executeUpdate() method is used to execute these operations and returns the number of records affected. JDBC provides a simple and efficient way to manage database records and is widely used in applications such as student management, banking, and inventory systems.

Exercise :

1. Create a Java application to manage **employee records** using CRUD operations.

```
import java.sql.*;

public class java20a {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jdbc_lab",
                "root",
                "anushree" );

            Statement st = con.createStatement();

            st.executeUpdate(
                "insert into employee values (101, 'Anushree', 'CSE',
50000),(102, 'Natasha', 'IT', 20000)"
            );

            ResultSet rs = st.executeQuery("select * from employee");

            System.out.println("Records inserted :");
            System.out.println("Employee Records");
            System.out.println("------------------------------");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("emp_id") + "  " +
                    rs.getString("name") + "  " +
                    rs.getString("department") + "  " +
                    rs.getInt("salary")
                );
            }
            System.out.println();

            st.executeUpdate(
                "update employee set salary = 55000 where emp_id = 102"
            );
```

```
            rs = st.executeQuery("select * from employee");

            System.out.println("Record updated :");
            System.out.println("Employee Records");
            System.out.println("------------------------------");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("emp_id") + "  " +
                    rs.getString("name") + "  " +
                    rs.getString("department") + "  " +
                    rs.getInt("salary")
                );
            }
            System.out.println();

            st.executeUpdate(
                "delete from employee where emp_id = 101"
            );

            rs = st.executeQuery("select * from employee");

            System.out.println("Record deleted :");
            System.out.println("Employee Records");
            System.out.println("------------------------------");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("emp_id") + "  " +
                    rs.getString("name") + "  " +
                    rs.getString("department") + "  " +
                    rs.getInt("salary")
                );
            }

            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

In SQL :

```
use jdbc_lab;
create table employee (
    emp_id int,
    name varchar(50),
    department varchar(50),
    salary double
);
```

OUTPUT :

```
Records inserted :
Employee Records
-----------------------------
101  Anushree  CSE  50000
102  Natasha  IT  20000

Record updated :
Employee Records
-----------------------------
101  Anushree  CSE  50000
102  Natasha  IT  55000

Record deleted :
Employee Records
-----------------------------
102  Natasha  IT  55000
PS C:\Users\softlink\Desktop\java code>
```

2. Create a Java application to manage **student records** using **CRUD operations**. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

```
import java.sql.*;

public class java20b {
    public static void main(String[] args) {

        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jdbc_lab",
                "root",
                "anushree" );

            Statement st = con.createStatement();

            // INSERT
            st.executeUpdate(
                "insert into student values " +
                "(101, 'Anushree', 'CSE', 93)," +
                "(102, 'Natasha', 'IT', 85)" );

            System.out.println("Records inserted");
            System.out.println("Student Records");
            System.out.println("------------------------------");

            ResultSet rs = st.executeQuery("select * from student");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("roll_no") + "  " +
                    rs.getString("name") + "  " +
                    rs.getString("course") + "  " +
                    rs.getInt("marks")
                );
            }

            // UPDATE
            st.executeUpdate(
                "update student set marks = 90 where roll_no = 102" );
```

```
            System.out.println("\nRecord updated");
            System.out.println("Student Records");
            System.out.println("------------------------------");

            rs = st.executeQuery("select * from student");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("roll_no") + "  " +
                    rs.getString("name") + "  " +
                    rs.getString("course") + "  " +
                    rs.getInt("marks")
                );
            }

            // DELETE
            st.executeUpdate(
                "delete from student where roll_no = 101" );

            System.out.println("\nRecord deleted");
            System.out.println("Student Records");
            System.out.println("------------------------------");

            rs = st.executeQuery("select * from student");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("roll_no") + "  " +
                    rs.getString("name") + "  " +
                    rs.getString("course") + "  " +
                    rs.getInt("marks")
                );
            }
            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

In SQL :

```
use jdbc_lab;
create table student (
    roll_no int,
    name varchar(50),
    course varchar(50),
    marks int
);
```

OUTPUT :

```
Records inserted
Student Records
-----------------------------
101  Anushree  CSE  93
102  Natasha  IT  85

Record updated
Student Records
-----------------------------
101  Anushree  CSE  93
102  Natasha  IT  90

Record deleted
Student Records
-----------------------------
102  Natasha  IT  90
PS C:\Users\softlink\Desktop\java code>
```

Outcome :

- Understood database modification operations using JDBC.
- Learned to insert, update, and delete records using SQL statements.
- Learned to retrieve records using the SELECT operation.
- Understood the use of executeUpdate() and ResultSet in JDBC.
- Successfully performed basic CRUD operations on database records.

Conclusion :

This assignment demonstrated database modification and retrieval operations using JDBC. Employee and student records were successfully inserted, retrieved, updated, and deleted using SQL statements. The assignment helped in understanding the basic implementation of CRUD operations in a Java application connected to a database.